Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program
(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**AI-Enabled Tactile System-on-Chip Integration for Soft Sensor Skins in Humanoids and Wearables**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical·Electronics		Mechanical·Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Devices and Systems	SoC, MEMS Element	Robotics/Automation Machinery	Measurement/Sensor Technologies for Automation

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Concept of Proposed Technology	This project proposes a tactile sensing and perception platform that combines soft wearable tactile sensors with on-device AI semiconductor chips to enable real-time tactile intelligence inside wearable devices and humanoid robots. The core concept is to move tactile data processing from external computing (PC/robot main CPU/cloud) to a low-power on-device AI chip integrated directly into the sensor module or robot limb. Soft tactile sensors (e.g., flexible sensor arrays/skins embedded in gloves, sleeves, or humanoid hands) produce rich, high-rate signals that are difficult to interpret using simple thresholding. We will develop AI algorithms that run on the chip to convert raw tactile signals into actionable events and states, such as: • Touch location and contact area estimation (where the contact occurs on the sensor/skin) • Touch type classification (tap, press, shear, slip onset, sustained contact) • Pressure/force pattern inference (distribution patterns correlated to grasp stability) • Interaction and manipulation state recognition (stable grasp vs. impending slip; object interaction signatures) Potential uses include: 1. Humanoid robot hands/skins for dexterous manipulation and safer human–robot interaction, and 2. Wearable soft sensing for assistive devices, teleoperation gloves, and ergonomic/rehabilitation monitoring where tactile-driven intent or interaction cues are needed. This joint R&D will deliver a hardware–sensor–algorithm co-designed solution: (i) chip integration and embedded inference pipeline, (ii) soft tactile sensor modules suitable for wearables/humanoids, and (iii) validated tactile AI models deployable on-device.

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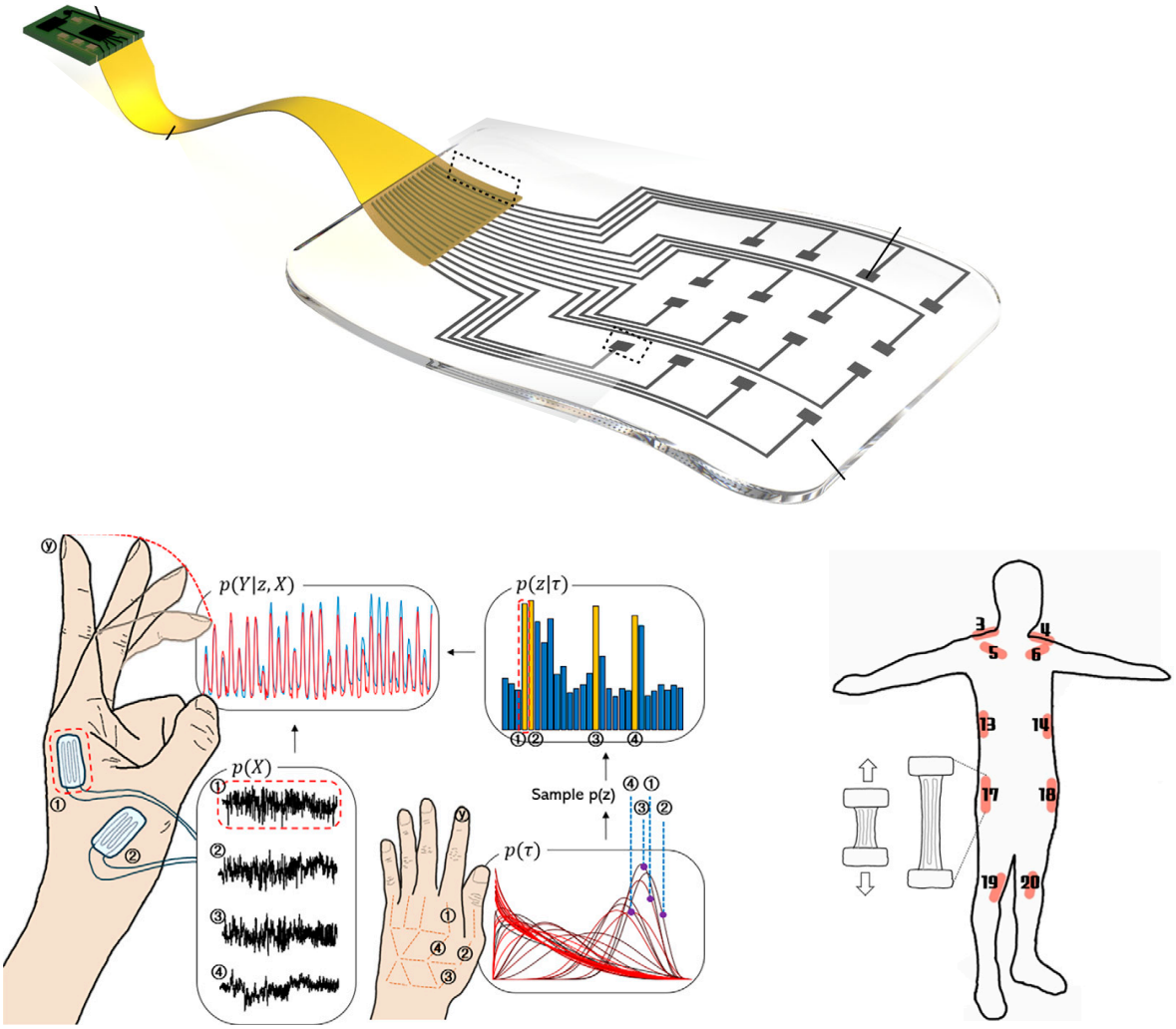

Figure. Project overview. A high-level illustration of the joint R&D concept: soft, conformable tactile sensor skins integrated with a low-power AI-enabled system-on-chip (SoC) to acquire and preprocess dense tactile signals, run on-device machine learning to convert raw sensor data into meaningful touch events/states, and enable real-time tactile intelligence for humanoid robots (and related wearable/robotic platforms). Images adapted from [2] and [3].

Rationale for Joint R&D Project	A joint R&D approach is required because delivering deployable tactile intelligence depends on tight co-optimization across three layers that no single party can efficiently complete alone. Under this arrangement, the Seoul National University (SNU) research team led by Prof. Yong-Lae Park will participate as a subcontractor under Korean SME, contributing soft-sensor design, fabrication, and integration tasks. The University of Toronto team will lead the tactile AI and system-level validation effort, co-led by Prof. Pakpong Chirarattananon (robotics, tactile-driven manipulation, humanoid/wearable application scenarios, and experimental evaluation) and Prof. Peyman Mohajerin Esfahani (AI and machine learning, robust/efficient learning under noisy, drifting tactile signals and hardware-

aware model optimization)

1. **Semiconductor / chip integration (Korean SME)**
   - Integrate/optimize on-device AI capability for tactile workloads (streaming sensor inputs, real-time inference, low latency).
   - Provide embedded processing, power management, and interfaces suitable for sensor modules and robotic platforms.
   - Leverage existing on-device AI chip to reduce time-to-prototype and manage funding constraints versus developing a new chip from scratch.
   - A **merged and optimized SoC variant** may be considered as a followon development after successful validation of the module concept within this project.
2. **Soft wearable tactile sensor development (SNU: Prof Yong-Lae Park)**
   - Develop robust, conformable tactile sensors/skins appropriate for humanoids and wearables (mechanically compliant, repeatable, scalable).
   - Define sensor layouts and packaging that are manufacturable and integrable with the chip module.
3. **Tactile AI algorithms (University of Toronto)**
   - Develop ML/AI methods tailored to tactile data characteristics (noise, drift, hysteresis, multi-modal signals) and constrained embedded deployment.
   - Create models for touch localization and touch-type recognition (e.g., slip/contact/pressure regimes) and interaction-state estimation.
   - Optimize models for on-device execution (latency, memory footprint, power) and provide deployment pipeline onto the partner's chip.

Collaboration is necessary to ensure the sensor design produces learnable, stable signals, the AI algorithms are hardware-aware, and the chip firmware/runtime supports real-time inference within practical power and size limits—enabling a credible path to industrialization for humanoid robotics and wearable pro

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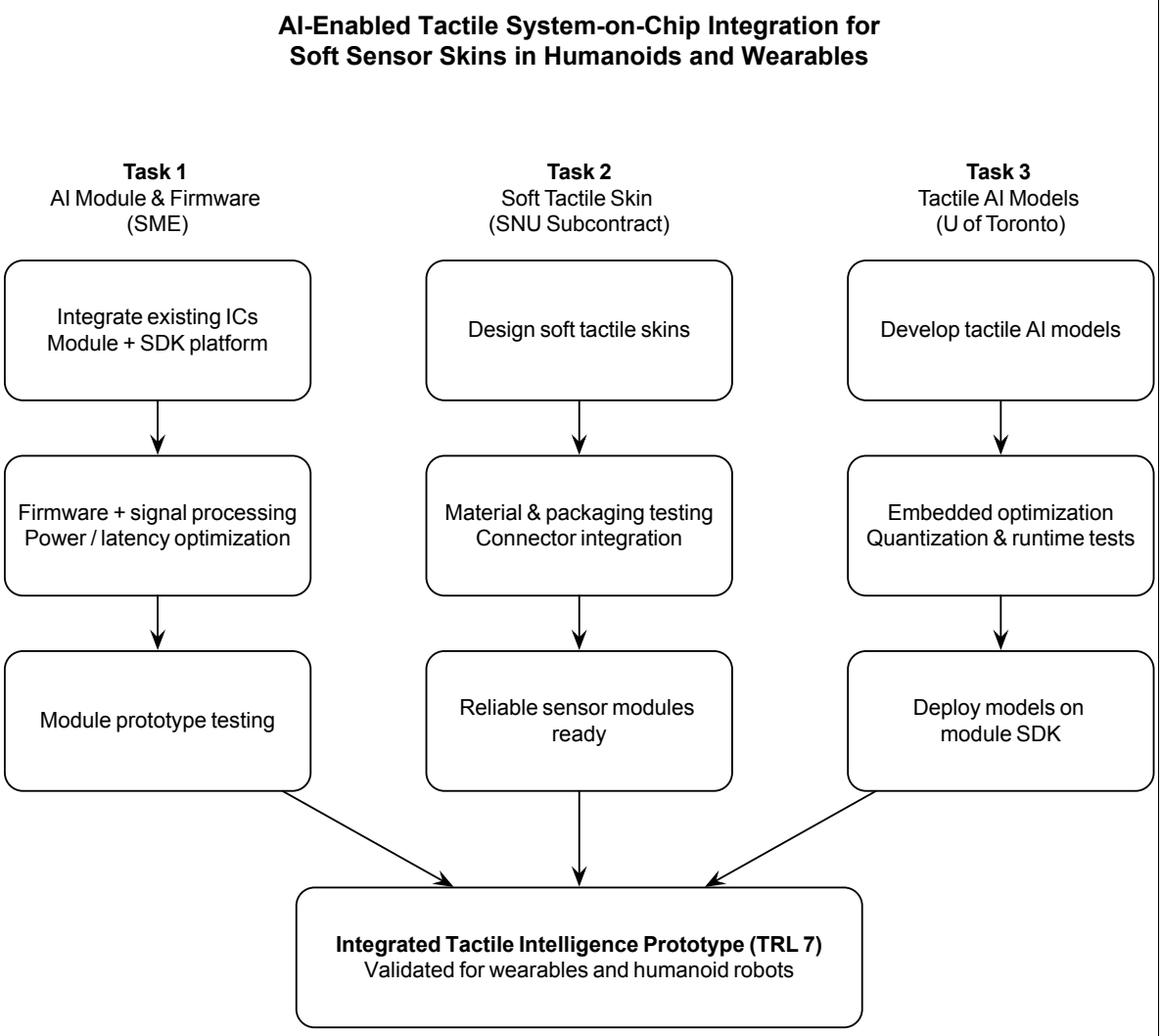
ducts.



Key Words	1. On-device AI 2. Soft wearable tactile sensors 3. Humanoid robotics

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	**Objective 1 — Semiconductor / SoC integration (Korean SME partner)** Develop and validate a **low-power tactile AI module and SDK integration**, leveraging **pre-developed ICs** (for sensor interface, AI acceleration, and data acquisition) to support continuous, real-time tactile sensing workloads. • Deliverables: sensor interface/readout integration plan, embedded runtime/firmware for streaming acquisition + inference, power/latency optimization, and a deployable on-device inference SDK running on the existing partner chip. **Objective 2 — Soft tactile sensor skins & packaging (SNU – Prof. Yong-Lae Park team)** Develop robust, conformable soft tactile sensor skins/modules for humanoids and wearables that produce stable, learnable signals and are manufacturable/integrable with the SoC module. • Deliverables: sensor skin prototypes (array layout, materials stack, wiring/routing, packaging), calibration procedures, and integration-ready sensor modules compatible with the SME electronics. **Objective 3 — Tactile AI algorithms & hardware-aware deployment (University of Toronto)** Develop tactile perception models (touch localization/contact area, touch-type classification, pressure/force pattern inference, slip/manipulation-state recognition) and optimize them for embedded on-device execution(latency/memory/power). • Deliverables: curated tactile datasets + labeling protocol, model architectures, quantized/optimized models, evaluation benchmarks, and deployment package integrated into the SME runtime.			
R&D Target Specifications	**1) SoC / Embedded integration (Korean SME)** • Real-time latency: ≤ 20 ms from sensor input to AI output • Supported sensor size: ≥ 64 channels (e.g., 8×8) at ≥ 200 Hz/taxel (or equivalent throughput) • Power: average ≤ 200 mW during continuous operation (target) • Memory: model + runtime ≤ 2 MB (target, quantized deployment) • Operation: core inference runs on-device (no PC/cloud required) **2) Soft tactile sensor skin/module (SNU)** • Conformable array prototype: ≥ 3×3 taxels on a flexible/soft patch • Durability: ≥ 100,000 contact cycles without failure (standard test to be defined) • Stability: ≤ 10% sensitivity change over 1 hour continuous use after calibration • Integrability: packaging/connector ready to interface with SME electronics **3) Tactile AI models & deployment (UofT)** • Touch localization: error ≤ 1 taxel (or ≤ 10 mm, depending on pitch) • Touch-type recognition: macro-F1 ≥ 0.85 (tap/press/shear/slip onset/sustained)			

	• Slip / manipulation-state detection: F1 ≥ 0.85 (and early warning ≥ 50 ms, where applicable) • On-device retention: ≥ 90% of offline accuracy after quantization and deployment
R&D Plan	This joint R&D is structured into three tasks aligned to the three objectives and participating organizations. The SNU team (subcontracted under Gwanak Analog) will execute Task 2 (soft-sensor skins and packaging) in coordination with the Korean SME (Task 1) and the University of Toronto (Task 3). Work begins with a 6-month preplanning phase (Jul–Dec 2026) to finalize scope, freeze interfaces, and establish joint evaluation metrics, followed by the main project (Jan 2027–Jun 2029) that progresses from initial integrated prototypes (TRL 4–5) to demonstration- and evaluation-ready prototypes (TRL 7). Jul–Dec 2026 (Preplanning, 6 months): Scope + interface freeze (all partners) • Define target use cases and metrics: humanoid manipulation and wearable interaction; outputs include touch localization/contact area, touch-type recognition, and slip/manipulation-state detection. Agree on datasets, test protocols, and acceptance metrics. • Freeze system interfaces (v0): sensor skin ↔ connector/packaging ↔ readout/SoC; sampling/throughput assumptions; data formats and calibration hooks. • Risk-reduction plan: select initial sensor variant(s), confirm feasibility on the target SoC, and design early test rigs for repeatable data collection and labeling. **Task 1 — SoC / Embedded Integration (Korean SME partner)** Goal: Real-time tactile acquisition + on-device inference pipeline using a module + SDK platform built from existing ICs, meeting power/latency constr aints for robot sensing applications. 1. Jan–Sep 2027 (Bring-up): implement sensor interface and streaming capture, buffering/timestamping, preprocessing hooks, and integrate a quantized-capable inference runtime. Deliver a module-level prototype (HW + FW + SDK) for lab testing. 2. Oct 2027–Dec 2028 (Optimization/iteration): co-optimize firmware scheduling and dataflow (DMA, batching/event options), reduce latency/power, and iterate module revisions as sensor packaging evolves (noise mitigation, connector robustness). 3. Jan–Jun 2029 (Hardening/evaluation): finalize stable firmware, error handling, and multi-unit build procedures; support integration into humanoid/wearable demos and partner/potential-customer evaluations (TRL 7). **Task 2 — Soft Tactile Sensor Skin/Module (SNU – Prof. Yong-Lae Park team)**

Goal: Manufacturable, conformable tactile skins/modules that produce stable, learnable signals and integrate with the SME electronics.

1. Jan–Sep 2027 (Sensor v1): design/fabricate initial soft tactile arrays/skins (e.g., 8×8 taxels or equivalent), define calibration, and characterize noise, drift/hysteresis, repeatability, and basic durability.
2. Oct 2027–Dec 2028 (Packaging + reliability): develop integration-ready packaging (routing, encapsulation, flex/connector, strain relief) and run durability/contact-cycle tests; iterate materials/layout based on signal quality and integration feedback.
3. Jan–Jun 2029 (Demo-ready modules): deliver integration-ready sensor modules with documented assembly/mounting/calibration procedures for multi-unit evaluation and demonstrations.

Task 3 — Tactile AI Algorithms & Hardware-Aware Deployment (University of Toronto)

Goal: Tactile perception models that run on-device and output actionable events/states.

1. Jan–Sep 2027 (Dataset v1 + baselines): collect and label representative tactile data (tap/press/shear, incipient slip, sustained contact) and train baseline models for touch localization and touch-type recognition; establish benchmarks that guide sensor and firmware choices.
2. Oct 2027–Dec 2028 (Robustness + embedded optimization): improve robustness to drift, unit variation, and mounting; quantize/compress models and simplify preprocessing for embedded execution; integrate into the SME runtime and iterate jointly to meet latency/memory/power targets, including slip/manipulation-state estimation.
3. Jan–Jun 2029 (Final models + release package): validate end-to-end performance on the target SoC with final sensor modules; lock model versions and deliver a reproducible training→deployment pipeline with test vectors and documentation to support TRL 7 evaluations.

Outcome: By Jun 2029, the consortium delivers integrated tactile sensor modules with on-device AI, validated in humanoid and wearable demonstrations and prepared for external evaluation consistent with TRL 7.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Define perception targets and metrics: specify outputs (touch localization/contact area, touch-type recognition, slip/manipulation-state detection) and success criteria aligned to humanoid and wearable demos. • Dataset + benchmarking: plan experiments, collect/label tactile datasets from integrated prototypes, and maintain evaluation benchmarks and test reports. • Model development: develop compact, robust tactile AI models that handle noise/drift/hysteresis and generalize across sessions and sensor units. • Embedded optimization support: deliver quantized/compressed models, reference implementations, test vectors, and deployment documentation compatible with the SME runtime constraints. • System-level technical coordination: run joint design reviews, support debugging of end-to-end performance issues (accuracy/latency), and ensure the algorithm requirements are reflected in sensor/electronics interfaces.
Expected R&R of Korean SME Partner	**Korean SME Partner (Task 1: SoC/embedded integration)** • Provide the on-device AI SoC platform: sensor interfaces, embedded runtime/firmware, buffering/preprocessing, power management, and real-time inference execution. • Electronics/module prototyping: design and build integration hardware (PCBs/modules, connectors, packaging support) to connect the tactile skin reliably to the SoC. • On-device deployment and optimization: integrate UofT models on-chip and optimize latency, memory, and power; maintain a repeatable build/deployment workflow and verification tests. • Validation and external evaluation support: lead multi-unit prototype build, stability/reliability testing, and coordinate partner/potential-customer evaluations toward TRL 7. **SNU Research Team (Task 2: Soft tactile sensor skin/module)** • Design and fabricate soft tactile skins/modules: develop conformable sensor arrays/skins suitable for humanoids and wearables. • Packaging and integratability: deliver integration-ready sensor packaging (routing, encapsulation, connectors/flex, strain relief) compatible with SME electronics constraints. • Characterization and calibration: quantify sensitivity, noise, drift, and durability; provide calibration procedures and integration guidelines to support reliable on-device learning/inference. • Iteration support: iterate sensor design based on measured system performance and feedback from SME integration and UofT model requirements. **Joint (SME + SNU) collaboration commitments** • Co-define sensor–electronics interfaces early, share design constraints promptly, and support rapid iteration (regular technical meetings, shared test data, integration troubleshooting).

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Key technical outcome:** an AI-enabled tactile module (soft sensor skin + embedded electronics) that performs real-time on-device tactile perception. **Demonstration outcome:** validated prototypes in two scenarios, humanoid manipulation (e.g., grasp/slip awareness) and wearable interaction sensing (touch events/types), with readiness for TRL 7-style partner/customer evaluation. **Commercialization potential:** a reusable component for humanoid robot hands/skins, teleoperation/assistive gloves, and automation end-effectors, with advantages in lower latency, lower bandwidth, and simpler system integration versus off-board processing. **Broader impact:** strengthens Korea's semiconductor + AI + robotics/manufacturing value chain, supports safer human–robot interaction, and creates follow-on opportunities for IP, products, and continued joint development.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization